

A decorative graphic on the left side of the slide consisting of two overlapping parallelograms. The front one is blue and the back one is a light green. They are positioned diagonally, with the blue one partially covering the green one.

Predicting Air Quality Outcomes Through Machine Learning and Environmental Feature Engineering

Daniel Buckholz
UMGC Data 690



Scope

Project Goal

- Analyze global air quality patterns using machine learning and data analytics techniques.
- Identify demographic, economic, and environmental factors associated with air quality outcomes.
- Integrate multiple public datasets to create a richer analytical framework.
- Develop predictive models capable of estimating national air quality levels.

Project Boundaries

- Country-level analysis of global air quality data.
- Global Air Pollution Dataset combined with World Bank Development Indicators.
- Focus on 2023 socioeconomic and environmental indicators.
- Emphasis on prediction and relationship discovery rather than causal determination.

Project Outcomes

- Identification of key factors influencing air quality.
- Actionable insights from data visualization and exploratory analysis.
- Evaluation of machine learning models for air quality prediction.
- Recommendations for future environmental analytics research.



Problem Definition

The Research Problem

- Air pollution remains a major environmental and public health challenge worldwide.
- Air quality is influenced by numerous demographic, economic, and environmental factors.
- Many studies focus on individual variables, but air quality is often driven by multiple interacting conditions.
- Understanding these relationships can support better environmental planning and policy decisions.

Research Question

Which demographic, economic, and environmental factors are most strongly associated with national air quality outcomes, and can the integration of multiple public datasets improve the prediction of air quality conditions?



Datasets

Global Air Pollution Dataset (Kaggle)

- 23,463 Observations
- 175 Countries
- Provides Target Variable (AQI)
- Pollutant AQI Variables
 - CO
 - O₃
 - NO₂
 - PM2.5

World Bank Indicators (World Bank)

- Population
- GDP Per Capita
- Industry Value Added (% GDP)
- Energy Use Per Capita
- Urban Population
- Greenhouse Gas Emissions Per Capita
- Provides Predictor Variables

Pollution Dataset
+
World Bank Indicators
↓
Integrated Country-level Dataset
(175 countries)

Data Preparation and Integration

Data Processing Steps

- Cleaned and standardized country names across datasets.
- Selected relevant World Bank indicators for analysis.
- Aggregated air quality measurements to the country level.
- Merged pollution and development datasets.
- Removed unmatched records and validated data quality.

Final Dataset

- 175 countries represented.
- Air quality metrics combined with demographic, economic, and environmental indicators.
- Dataset prepared for visualization and machine learning analysis.

Challenges Encountered

- Country naming inconsistencies between sources.
- Missing values for selected indicators.
- Variable selection and feature engineering decisions.
- Data quality validation after merging.



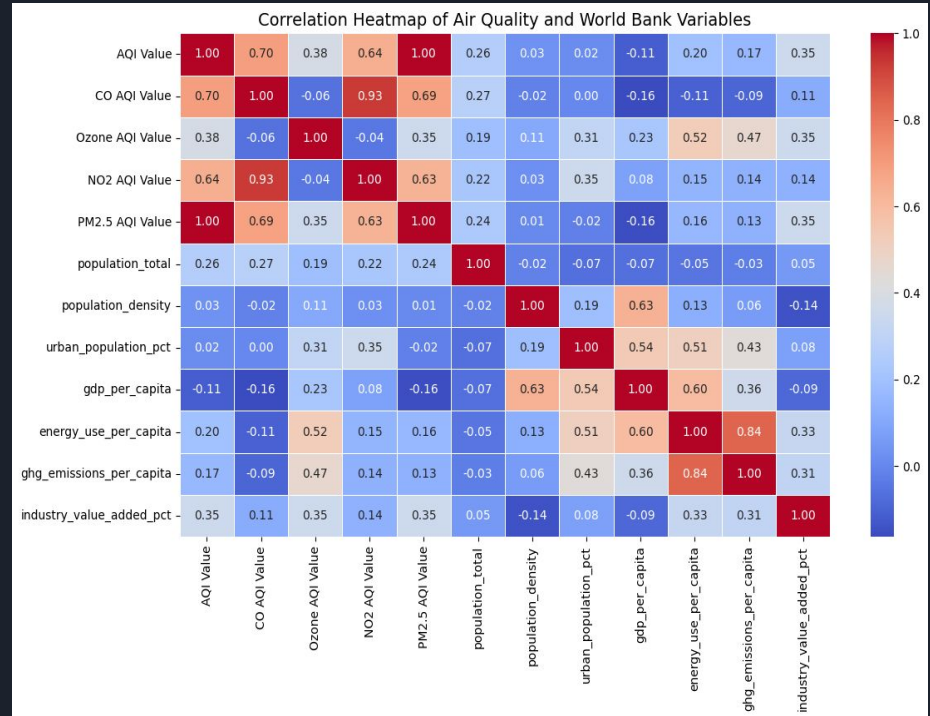
Correlation Heatmap

Purpose

- Examine relationships among air quality, demographic, economic, and environmental variables.
- Identify patterns that may support predictive modeling.
- Detect potential multicollinearity among features.

Key Findings

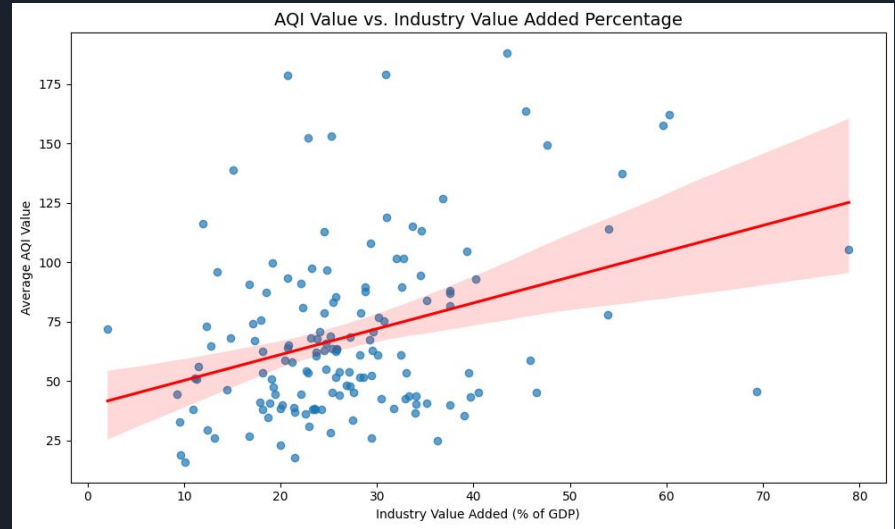
- AQI shows relationships with several development indicators.
- Economic and environmental variables exhibit moderate correlations.
- No single variable fully explains air quality outcomes.
- Multiple factors appear to contribute to national AQI levels.



AQI vs. Industry Value Added (%)

Key Findings

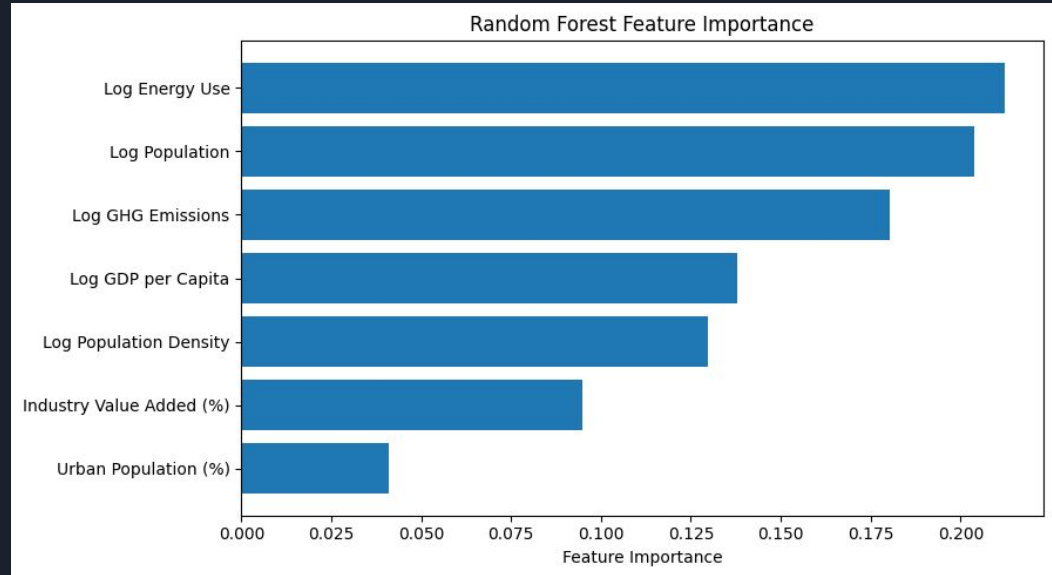
- Positive overall trend.
- High variability across countries.
- Industry appears influential.
- Additional factors likely contribute.



Random Forest Feature Importance

Key Findings

- Log Energy Use was the strongest predictor of national average AQI.
- Log Population and Log GHG Emissions were the next most influential features.
- Log GDP per Capita and Log Population Density also contributed meaningful predictive value.
- Industry Value Added remained informative but ranked below several transformed variables.
- The model indicates that air quality is influenced by multiple interacting socioeconomic and environmental factors rather than any single predictor.



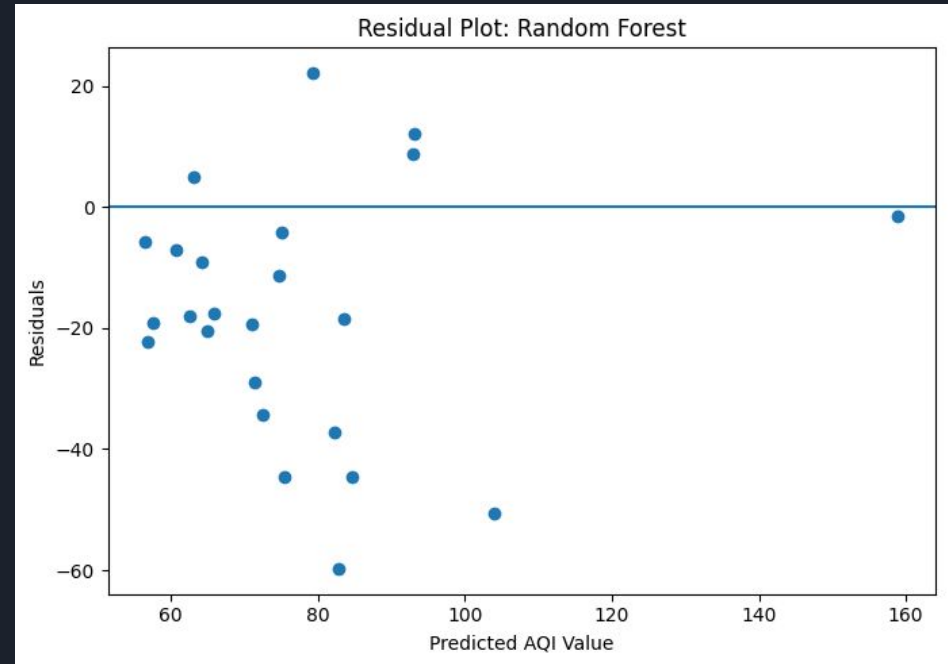
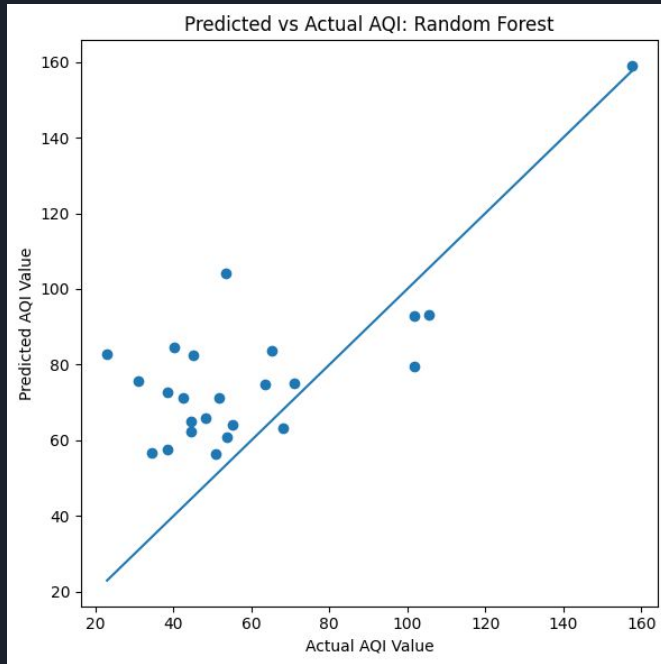


Predictive Models

Model	CV R^2	Test R^2	MAE	RMSE
Multiple Linear Regression	—	0.060	23.32	28.58
Random Forest	0.366	0.175	21.76	26.76
Gradient Boosting	0.380	0.084	22.36	28.21

Champion Model: Random Forest

Champion Model Performance





Key Findings

- Air quality is influenced by multiple interacting socioeconomic and environmental factors.
- Energy Use, Population, and GHG Emissions were the strongest predictors.
- Random Forest produced the strongest predictive performance.
- Socioeconomic variables explained part (but not all) of the variation in AQI.
- Additional meteorological and geographic variables are likely needed to improve prediction.



KPI Performance

KPI	Result
Successful dataset integration	✓
Three ML models completed	✓
Champion model identified	✓
Visualizations created	✓
Research question addressed	✓



Lessons Learned

Lesson 1 - Data Integration

Data integration is often the hardest part of analytics.

Lesson 2 - Feature Engineering

Feature engineering can have as much impact on predictive performance as algorithm selection.

Lesson 3 - Complex Systems

Environmental systems are shaped by multiple interacting socioeconomic and environmental factors.

Lesson 4 - Analytical Insight

Unexpected results often provide the most valuable analytical insights.



Future Research

- Add meteorological variables
- Temporal forecasting
- Satellite data
- Policy indicators
- Geographic modeling
- Deep learning approaches



Conclusion

Research Question

- Can demographic, economic, and environmental variables be integrated to improve the prediction of national air quality?

Approach

- Integrated air pollution and World Bank datasets, engineered predictive features, and evaluated three machine learning models.

Outcome

- Random Forest produced the strongest predictive performance, demonstrating that national air quality is driven by multiple interacting socioeconomic and environmental factors.



Integrating multiple public datasets with machine learning provided meaningful insight into the complex factors associated with national air quality.



References

Breiman, L. (2001). *Random forests*. *Machine Learning*, 45(1), 5–32. <https://doi.org/10.1023/A:1010933404324>

Byrne, G. (2007). *A statistical primer: Understanding descriptive and inferential statistics*. *Evidence Based Library and Information Practice*, 2(1), 32–47. <https://doi.org/10.18438/B8FW2H>

Cleveland, W. S., & McGill, R. (1984). *Graphical perception: Theory, experimentation, and application to the development of graphical methods*. *Journal of the American Statistical Association*, 79(387), 531–554. <https://doi.org/10.1080/01621459.1984.10478080>

Fernandez, N. F., Gundersen, G. W., Rahman, A., Grimes, M. L., Rikova, K., Hornbeck, P., & Ma'ayan, A. (2017). *Clustergrammer, a web-based heatmap visualization and analysis tool for high-dimensional biological data*. *Scientific Data*, 4(1), 170151. <https://doi.org/10.1038/sdata.2017.151>

Friedman, J. H. (2001). *Greedy function approximation: A gradient boosting machine*. *The Annals of Statistics*, 29(5), 1189–1232. <https://doi.org/10.1214/aos/1013203451>

Gupta, N. S., Mohta, Y., Heda, K., Armaan, R., Valarmathi, B., & Arulkumaran, G. (2023). *Prediction of air quality index using machine learning techniques: A comparative analysis*. *Journal of Environmental and Public Health*, 2023 <https://doi.org/10.1155/2023/4916267>

James, G., Witten, D., Hastie, T., & Tibshirani, R. (2021). *An introduction to statistical learning: With applications in R* (2nd ed.). Springer. <https://doi.org/10.1007/978-1-0716-1418-1>

Jobin, A., Ienca, M., & Vayena, E. (2019). *The global landscape of AI ethics guidelines*. *Nature Machine Intelligence*, 1(9), 389–399. <https://doi.org/10.1038/s42256-019-0088-2>

Kaneko, H. (2023). *Interpretation of machine learning models for data sets with many features using feature importance*. *ACS Omega*, 8(25), 23218–23225. <https://doi.org/10.1021/acsomega.3c03722>

Muzdadid, H. A. (2023). *Global air pollution dataset* [Data set]. Kaggle. <https://www.kaggle.com/datasets/hasibalmuzdadid/global-air-pollution-dataset/data>

Rybarczyk, Y., & Zalakeviciute, R. (2018). *Machine learning approaches for outdoor air quality modelling: A systematic review*. *Applied Sciences*, 8(12), 2570. <https://doi.org/10.3390/app8122570>

World Bank Group. (2023). *World development indicators: Population, total (SP.POP.TOTL)*. <https://databank.worldbank.org/source/world-development-indicators>

Zhang, Y., Bocquet, M., Mallet, V., Seigneur, C., & Baklanov, A. (2012). *Real-time air quality forecasting, part I: History, techniques, and current status*. *Atmospheric Environment*, 60, 632–655. <https://doi.org/10.1016/j.atmosenv.2012.06.031>